

Final Exam

Problems

Exercise 1 (1 point).

In regression analysis, which of the following is **not** a required assumption about the error term ε ?

- A. The expected value of the error term is zero.
 - B. The variance of the error term is the same for all values of X .
 - C. The values of the error term are dependent.
 - D. The error term is normally distributed.
-

Exercise 2 (1 point).

In regression analysis, the response variable is the

- A. dependent variable
 - B. independent variable
 - C. slope of the regression function
 - D. intercept of the regression function
-

Exercise 3 (1 point).

A variable that takes on the values of 0 or 1 and is used to incorporate the effect of categorical variables in a regression model is called

- A. a constant variable
 - B. an interaction
 - C. a dummy variable
 - D. None of these alternative is correct.
-

Exercise 4 (1 point).

If a categorical variable has k levels, ____ indicator variables are required with each indicator being coded as 0 or 1.

- A. k
- B. $k - 1$
- C. $k - 2$
- D. None of these alternatives is correct.

Exercise 5 (1 point).

The following regression model is known as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \varepsilon.$$

- A. Sixth order model with one predictor
 - B. First order model with one predictor
 - C. First order model with six predictors
 - D. Sixth order model with six predictors
-

Exercise 6 (1 point).

Which of the following tests is used to determine whether an additional set of several independent variables makes a significant reduction in SSE to a multiple regression model?

- A. Z-test
 - B. t-test
 - C. Overall F-test
 - D. Partial F-test
-

Exercise 7 (1 point).

A variable $x_1 x_2$ is added to a multiple regression model in order to account for potential effects of two variables x_1 and x_2 acting together. This type of effect is

- A. called multicollinearity effect
 - B. called transformation effect
 - C. called interaction
 - D. called quadratic term
-

Exercise 8 (1 point).

The variable selection procedure that selects the best regression model, for a given number of independent variables, is known as

- A. backwards elimination
- B. forward selection
- C. stepwise regression
- D. all-possible-regression selecton (aka. best subsets regression)

Rain Level part 1

The state of California operates numerous meteorological stations. One of the many functions of each station is to monitor rainfall on a daily basis. This information is then used to produce an average annual precipitation level for each station.

Station	Precip_in	Altitude_ft	Latitude_deg	Distance_miles	Shadow
Eureka	39.57	43	40.8	1	W
Red Bluff	23.27	341	40.2	97	L
Thermal	18.20	4152	33.8	70	L
Fort Bragg	37.48	74	39.4	1	W
Soda Springs	49.26	6752	39.3	150	W
San Francisco	21.82	52	37.8	5	W
Sacramento	18.07	25	38.5	80	L
San Jose	14.17	95	37.4	28	L
Giant Forest	42.63	6360	36.6	145	W
Salinas	13.85	74	36.7	12	L
Fresno	9.44	331	36.7	114	L
Ft. Piedras	19.33	57	35.7	1	W
Paso Robles	15.67	740	35.7	31	L
Bakersfield	6.00	489	35.4	75	L
Bishop	5.73	4108	37.3	198	L
Mineral	47.82	4850	40.4	142	W
Santa Barbara	17.95	120	34.4	1	W
Susanville	18.20	4152	40.3	198	L
Tule Lake	10.03	4036	41.9	140	L
Needles	4.63	913	34.8	192	L
Burbank	14.74	699	34.2	47	W
Los Angeles	15.02	312	34.1	16	W
Long Beach	12.36	50	33.8	12	W
Los Banos	8.26	125	37.8	74	L
Blythe	4.05	268	33.6	155	L
San Diego	9.94	19	32.7	5	W
Daggett	4.25	2105	34.1	85	L
Death Valley	1.66	-178	36.5	194	L
Crescent City	74.87	35	41.7	1	W
Colusa	15.95	60	39.2	91	L

Table above lists average annual precipitation levels (in inches) for a sample of 30 meteorological stations scattered throughout the state. In addition to average annual precipitation (y), the table lists three independent variables that are believed (by California geographers) to have the most impact on the amount of rainfall at each station, as follows:

- Altitude of the station (x_1 , feet)
- Latitude of the station (x_2 , degrees)
- Distance of the station from the Pacific coast (x_3 , miles)

As an initial attempt in explaining the average annual precipitation in california, we consider the following simple linear regression model:

$$E(y) = \beta_0 + \beta_2 x_2$$

where x_2 is the latitude of the station. The R printout is as follows:

Analysis of Variance Table

Response: Precip_in

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Latitude_deg	1	2664.9	2664.89	13.956	0.0008495 ***
Residuals	28	5346.8	2.____		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Call:

lm(formula = Precip_in ~ Latitude_deg)

Residuals:

Min	1Q	Median	3Q	Max
-27.297	-7.956	-2.103	6.082	38.262

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-113.3028	35.7210	-3.172	0.00366 **
Latitude_deg	3.5950	0.9623	3.____	0.00085 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.____ on 28 degrees of freedom

Multiple R-squared: 0.3326, Adjusted R-squared: 0.3088

F-statistic: 13.96 on 1 and 28 DF, p-value: 0.0008495

Exercise 9 (1 point).

Please fill in the value of 1.____.

Exercise 10 (1 point).

Please fill in the value of 2.____.

Exercise 11 (1 point).

Please fill in the value of 3.____.

Exercise 12 (1 point).

Please fill in the value of 4.____.

Exercise 13 (1 point).

Please interpret the coefficient of determination equation R^2 .

Rain Level part 2

Next, we consider using more predictors to predict the average annual precipitation. We consider the following first-order regression model:

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3.$$

Call:

```
lm(formula = Precip_in ~ Altitude_ft + Latitude_deg + Distance_miles)
```

Residuals:

Min	1Q	Median	3Q	Max
-28.722	-5.603	-0.531	3.510	33.317

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.024e+02	2.921e+01	-3.505	0.001676 **
Altitude_ft	4.091e-03	1.218e-03	3.358	0.002431 **
Latitude_deg	3.451e+00	7.949e-01	4.342	0.000191 ***
Distance_miles	-1.429e-01	3.634e-02	-3.931	0.000559 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.1 on 26 degrees of freedom

Multiple R-squared: 0.6003, Adjusted R-squared: 0.5542

F-statistic: 13.02 on 3 and 26 DF, p-value: 2.205e-05

Analysis of Variance Table

	Df	SS	MS	F	P
Source	3	4809.4	1603.12	13.016	2.2051e-05
Error	26	3202.3	123.17		
Total	29	8011.7			

Exercise 14 (1 point).

Fit the model to the data using the method of least squares. (Round the coefficient to three decimal places.)

Exercise 15 (1 point).

Test whether the overall model is statistically useful for predicting the average annual precipitation using $\alpha = .05$. What is the null and alternative hypothesis?

- A. $H_0 : \beta_0 = \beta_1 = \beta_2 = \beta_3 = 0, H_a : \beta_0 \neq \beta_1 \neq \beta_2 \neq \beta_3 \neq 0$.
- B. $H_0 : \beta_0 = \beta_1 = \beta_2 = \beta_3 = 0, H_a$: At least one of the β 's $\neq 0$.
- C. $H_0 : \beta_1 = \beta_2 = \beta_3 = 0, H_a$: At least one of the β 's $\neq 0$.
- D. $H_0 : \beta_1 = \beta_3 = 0, H_a : \beta_0 \neq \beta_1 \neq \beta_2 \neq \beta_3 \neq 0$.

Exercise 16 (1 point).

Test whether the overall model is statistically useful for predicting the average annual precipitation using $\alpha = 0.05$. What is your conclusion?

Exercise 17 (1 point).

Give practical interpretations of $\hat{\beta}_2$.

Exercise 18 (1 point).

Find the model standard deviation, s , and interpret its value.

Rain level part 3

The researchers found that the stations located on the west-facing slopes of the California mountains invariably has higher average annual precipitation values than the leeward side of the mountains. To account for the shadow effect of the California mountains, we consider adding the dummy variable to the model.

The model with the shadow effect takes the form

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

where x_1 , x_2 , and x_3 are defined as before, and

$$x_4 = \begin{cases} 1 & \text{if station on the leeward side} \\ 0 & \text{if station on the westward side} \end{cases}$$

The R printout for the model is shown below.

Call:

```
lm(formula = Precip_in ~ Altitude_ft + Latitude_deg + Distance_miles +  
    Shadow)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.3157	-4.8023	0.0998	3.1054	28.7233

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.138e+02	2.431e+01	-4.679	8.56e-05 ***
Altitude_ft	2.205e-03	1.132e-03	1.948	0.06271 .
Latitude_deg	3.454e+00	6.561e-01	5.264	1.88e-05 ***
Distance_miles	-5.365e-02	3.879e-02	-1.383	0.17886
ShadowW	1.586e+01	4.371e+00	3.628	0.00128 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.16 on 25 degrees of freedom

Multiple R-squared: 0.7382, Adjusted R-squared: 0.6963

F-statistic: 17.62 on 4 and 25 DF, p-value: 5.437e-07

```
anova(fit_mlr, fit_dummy)
```

Analysis of Variance Table

Model 1: Precip_in ~ Altitude_ft + Latitude_deg + Distance_miles

Model 2: Precip_in ~ Altitude_ft + Latitude_deg + Distance_miles + Shadow

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	26	3202.3				
2	25	2097.8	1	1104.5	13.162	0.001279 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Exercise 19 (1 point).

Interpret the slope related with Shadow effect $\hat{\beta}_4$

Exercise 20 (1 point).

Does the shadow effect (x_4) contribute significantly in explaining the information in average annual precipitation (y)? Give the null and alternative hypothesis.

- A. $H_0 : \beta_0 = \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0, H_a : \text{At least one of the } \beta\text{'s} \neq 0.$
 - B. $H_0 : \beta_1 = \beta_2 = \beta_3 = 0, H_a : \text{At least one of the } \beta\text{'s} \neq 0.$
 - C. $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0, H_a : \beta_1 \neq \beta_2 \neq \beta_3 \neq \beta_4 \neq 0.$
 - D. $H_0 : \beta_4 = 0, H_a : \beta_4 \neq 0.$
-

Exercise 21 (1 point).

Conduct a test ($\alpha = 0.05$) to determine whether the shadow effect (x_4) contributes significantly in explaining the information in average annual precipitation (y). State the conclusion of the test in the words of the problem.

Rain Level part 4

The model with shadow effect is

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

Can the model above be improved by adding interaction terms? Consider the interaction model as follows:

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_1 x_4 + \beta_6 x_2 x_4 + \beta_7 x_3 x_4.$$

```
anova_alt(fit_dummy)
```

Analysis of Variance Table

	Df	SS	MS	F	P
Source	4	5913.8	1478.45	17.619	5.437e-07
Error	25	2097.8	83.91		
Total	29	8011.7			

```
anova_alt(fit_int)
```

Analysis of Variance Table

	Df	SS	MS	F	P
Source	7	6921.6	988.81	19.957	3.69e-08
Error	22	1090.0	49.55		
Total	29	8011.7			

```
anova(fit_dummy, fit_int)
```

Analysis of Variance Table

```
Model 1: Precip_in ~ Altitude_ft + Latitude_deg + Distance_miles + Shadow
Model 2: Precip_in ~ Altitude_ft + Latitude_deg + Distance_miles + Shadow +
  Shadow:Altitude_ft + Shadow:Latitude_deg + Shadow:Distance_miles
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1      25 2097.8
2      22 1090.0  3    1007.8 6.7804 0.002085 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Exercise 22 (1 point).

Conduct a hypothesis testing to determine whether these interaction terms are important. Give the null and alternative hypothesis.

- A. $H_0 : \beta_0 = \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = \beta_7 = 0$, H_a : At least one of the β 's $\neq 0$.
- B. $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = \beta_7 = 0$, H_a : At least one of the β 's $\neq 0$.
- C. $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$, H_a : At least one of the β 's $\neq 0$.
- D. $H_0 : \beta_5 = \beta_6 = \beta_7 = 0$, H_a : At least one of the β 's $\neq 0$.

Exercise 23 (1 point).

Compare the two above models. Which one is the reduced model?

A. $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$

B. $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_1 x_4 + \beta_6 x_2 x_4 + \beta_7 x_3 x_4$.

Exercise 24 (1 point).

Based on the R printout above, what is the SSE_R ? That is, the SSE of the reduced model.

Exercise 25 (1 point).

Based on the R printout above, what is the SSE_F ? That is, the SSE of the full model.

Exercise 26 (1 point).

Which one is the correct formula to calculate the F-statistics?

A. $F = \frac{\frac{2097.8 - 1090.0}{3}}{\frac{1090.0}{22}}$

B. $F = \frac{\frac{1090.0 - 2097.8}{7}}{\frac{1090.0}{22}}$

C. $F = \frac{\frac{2097.8 - 1090.0}{4}}{\frac{1090.0}{25}}$

D. $F = \frac{\frac{2097.8 - 1090.0}{3}}{\frac{1090.0}{25}}$

Exercise 27 (1 point).

Based on the R printout above, use partial F-test ($\alpha = 0.05$) to determine which model, the complete or reduced, is preferred. State the conclusion of the test in the words of the problem.

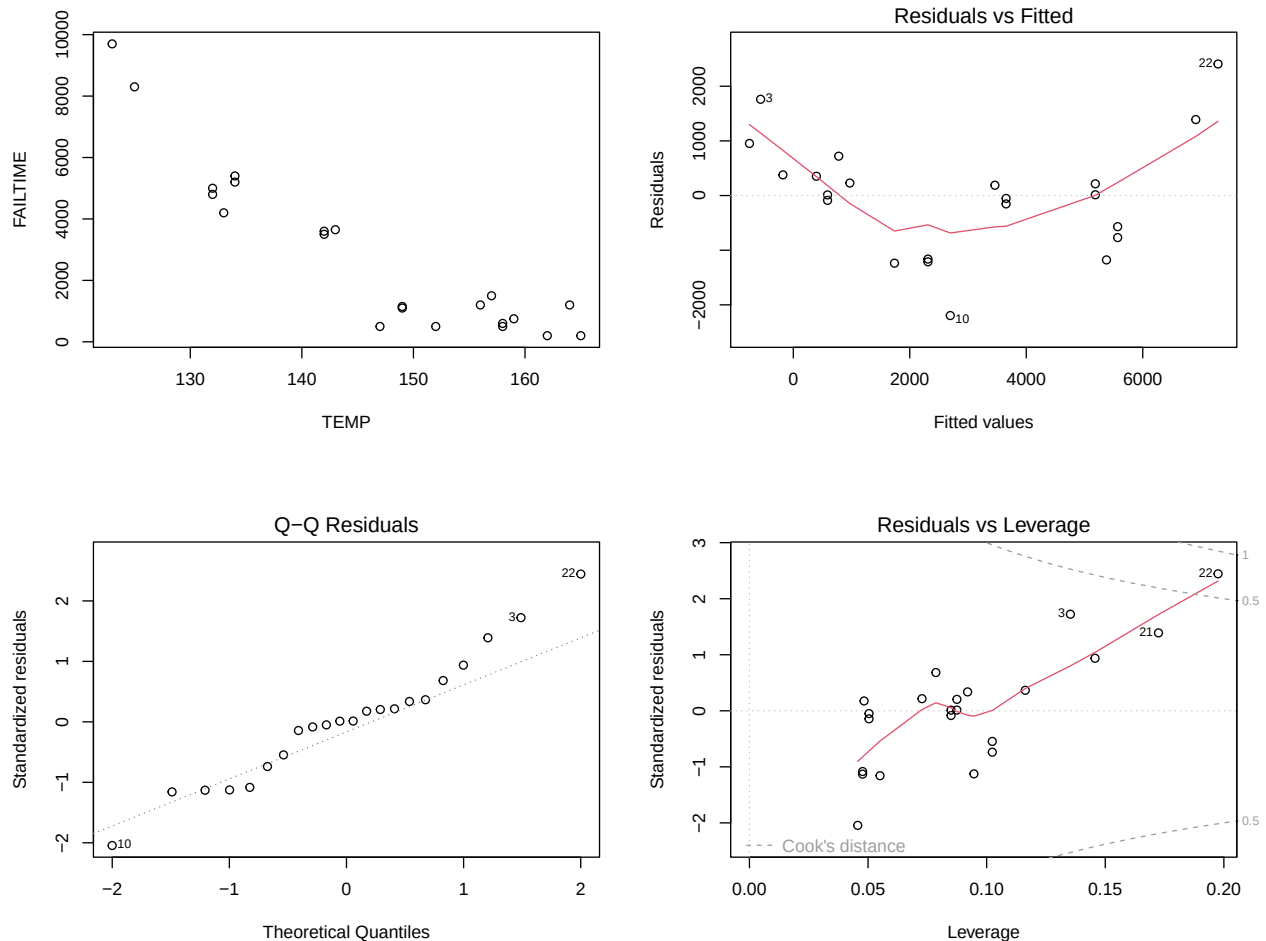
Semiconductor part 1

The national semiconductor study on using tinlead solder bumps to manufacture silicon wafer integrated circuit chips. The failure times of 22 microchips (in hours) were determined at different solder temperatures (degrees Centigrade), with the data repeated in the accompanying table. Recall that the researchers want to predict failure time (y) based on solder temperature (x).

Fit the straight-line model,

$$E(y) = \beta_0 + \beta_1 x,$$

to the data and plot the residuals against solder temperature.



Exercise 28 (1 point).

Based on the scatter plot and residual plot, do you detect a lack of fit?

Exercise 29 (1 point).

Comment on the normal QQ plot. Is the normality assumption reasonably satisfied?

Exercise 30 (1 point).

Comment on the residual vs leverage plot. Is there any influential observation?

Semiconductor part 2

The national semiconductor study on using tinlead solder bumps to manufacture silicon wafer integrated circuit chips. The failure times of 22 microchips (in hours) were determined at different solder temperatures (degrees Centigrade), with the data repeated in the accompanying table. Recall that the researchers want to predict failure time (y) based on solder temperature (x).

Fit the quadratic model

$$E(y) = \beta_0 + \beta_1 x + \beta_2 x^2.$$

Based on the output below, conduct a test to determine if there is upward curvature in the relationship between failure time and solder temperature. (Use $\alpha = 0.05$)

```
fit2 <- lm(FAILTIME ~ TEMP + I(TEMP^2))
summary(fit2)
```

Call:

```
lm(formula = FAILTIME ~ TEMP + I(TEMP^2))
```

Residuals:

Min	1Q	Median	3Q	Max
-1260.49	-475.70	-15.57	528.45	1131.69

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	154242.914	21868.474	7.053	1.03e-06 ***
TEMP	-1908.850	303.664	-6.286	4.92e-06 ***
I(TEMP^2)	5.929	1.048	5.659	1.86e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 688.1 on 19 degrees of freedom

Multiple R-squared: 0.9415, Adjusted R-squared: 0.9354

F-statistic: 152.9 on 2 and 19 DF, p-value: 1.937e-12

```
anova_alt(fit2)
```

Analysis of Variance Table

	Df	SS	MS	F	P
Source	2	144830280	72415140	152.93	1.9371e-12
Error	19	8997107	473532		
Total	21	153827386			

Exercise 31 (1 point).

What is the null hypothesis?

A. $H_0: \beta_1 = \beta_2 = 0$

B. $H_0: \beta_1 = 0$

C. $H_0: \beta_2 = 0$

D. $H_0: \beta_0 = 0$

Exercise 32 (1 point).

What is your conclusion?

Exercise 33 (1 point).

Interpret the estimated slope $\hat{\beta}_2 = 5.929$.

Variable selections

Exercise 34 (1 point).

Industrial engineers at the University of Florida used regression modeling as a tool to reduce the time and cost associated with developing new metallic alloys. To illustrate, the engineers built a regression model for the tensile yield strength (y) of a new steel alloy. The potential important predictors of yield strength are listed below.

- x_1 = Carbon amount (% weight)
- x_2 = Manganese amount (% weight)
- x_3 = Chromium amount (% weight)
- x_4 = Nickel amount (% weight)
- x_5 = Molybdenum amount (% weight)
- x_6 = Copper amount (% weight)
- x_7 = Nitrogen amount (% weight)
- x_8 = Vanadium amount (% weight)
- x_9 = Plate thickness (millimeters)
- x_{10} = Solution treating (milliliters)
- x_{11} = Aging temperature (degrees Celsius)

The stepwise regression selected the following independent variables: $x_1, x_2, x_3, x_5, x_6, x_8, x_9, x_{10}, x_{11}$. All these variables were statistically significant in the stepwise model, with $R^2 = .94$. Consequently, the engineers used the estimated stepwise model to predict yield strength. Do you agree with this decision? Explain.

Exercise 35 (1 point).

Based on the output below, apply a variable screening method to the data to find a small subset of the independent variables that are the best predictors of response variable MTBE level.

	(Intercept)	pH	DissOxy	IndPct	WellClassPublic	AquiferUnconsoli	Depth	Distance
1	*			*				
2	*			*			*	
3	*			*			*	*
4	*		*	*			*	*
5	*		*	*		*	*	*
6	*		*	*	*	*	*	*
7	*	*	*	*	*	*	*	*

Vars	R.sq	R.sq..adj.	Mallows.Cp
1	0.2007434	0.1965145	0.7862062
2	0.2101659	0.2017635	0.5723703
3	0.2137800	0.2011668	1.7232440
4	0.2165614	0.1997132	3.0697604
5	0.2193018	0.1982018	4.4259024
6	0.2205632	0.1951468	6.1295148
7	0.2211145	0.1913210	8.0000000

Please specify how you used the criteria (e.g. adjusted-R-sq, Cp) to draw the conclusion.

Quasar

A quasar is a distance celestial object (at least 4 billion light-years away) that provides a powerful source of radio energy. The data for a sample of 25 large (redshift) quasars is from a survey study of quasars detected by a deep space survey. The survey enabled astronomers to measure the rest frame equivalent width (the response variable y), redshift range (x_1), line flux (x_2), line luminosity (x_3), AB1450 magnitude (x_4) in a linear model below

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4.$$

A portion of the R spreadsheet showing the diagnostic table as follows.

RFEWIDTH	Predicted_Value	Residual	Std_Residual	Del_Residual	leverage_values	CooksD
117	136.7572	-19.7572	-1.4478	-1.4914	0.2163	0.1157
82	92.7556	-10.7556	-0.7204	-0.7114	0.0619	0.0069
33	0.9725	32.0275	2.4116	2.7912	0.2578	0.4041
92	101.5036	-9.5036	-0.7816	-0.7737	0.3779	0.0742
114	124.6313	-10.6313	-0.7326	-0.7239	0.1139	0.0138
50	43.6241	6.3759	0.4713	0.4620	0.2299	0.0133
43	34.3085	8.6915	0.5981	0.5882	0.1113	0.0090
259	210.3720	48.6280	4.1220	10.3582	0.4144	2.4043
58	57.2952	0.7048	0.0506	0.0493	0.1827	0.0001
126	126.9983	-0.9983	-0.0714	-0.0696	0.1777	0.0002
42	42.4372	-0.4372	-0.0342	-0.0333	0.3120	0.0001
146	144.9690	1.0310	0.0862	0.0840	0.3974	0.0010
124	122.4979	1.5021	0.1102	0.1075	0.2187	0.0007
75	78.0046	-3.0046	-0.2033	-0.1984	0.0808	0.0007
85	96.3953	-11.3953	-0.7649	-0.7567	0.0661	0.0083
109	115.8372	-6.8372	-0.5189	-0.5092	0.2694	0.0199
55	61.0258	-6.0258	-0.4192	-0.4104	0.1307	0.0053
91	94.8067	-3.8067	-0.2699	-0.2635	0.1627	0.0028
116	125.0699	-9.0699	-0.6369	-0.6272	0.1467	0.0140
75	77.2503	-2.2503	-0.1650	-0.1610	0.2177	0.0015
63	61.2439	1.7561	0.1200	0.1170	0.0982	0.0003
46	37.9654	8.0346	0.5933	0.5835	0.2283	0.0208
55	62.3593	-7.3593	-0.5316	-0.5218	0.1934	0.0136
99	108.0395	-9.0395	-0.6772	-0.6677	0.2502	0.0306
53	50.8796	2.1204	0.1437	0.1401	0.0835	0.0004

Exercise 36 (1 point).

Do you detect any outliers by using the standardized residual method? Explain.

Exercise 37 (1 point).

Do you detect any outliers by using the studentized deleted residual method? Explain.

Exercise 38 (1 point).

What is the critical value of cook's distance in R?

A. $qf(0.5, 5, 20) = 0.9004$

C. $qf(0.05, 5, 20) = 0.2193$

B. $qf(0.5, 4, 21) = 0.8669$

D. $qf(0.05, 4, 21) = 0.1726$

Exercise 39 (1 point).

Do you detect any influential observations by using Cook's distance method? Explain.